

(M19IT1101)
SAGIRAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)
I M.Tech I SEMESTER (R19) Regular Examinations
DISCRETE MATHEMATICAL STRUCTURES
Department of Information Technology

Time: 3 Hours

Max. Marks 75

Answer ONE question from EACH UNIT

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1	a)	Solve for the value of c, distribution function of X and $P(X \geq 3)$, given $f(x) = \frac{c}{3^x}$ for $x = 1, 2, 3, \dots, n$ as the probability function of the random variable X.	CO1	K3	7
	b)	The joint probability function of two discrete random variables X and Y is given by $f(x, y) = c(2x + y)$ where X and Y can assume all integers such that $0 \leq x \leq 2$, $0 \leq y \leq 3$ and $f(x, y) = 0$ other wise. Solve for i) the value of c ii) E(X) iii) E(Y) iv) Var(X) and Var(Y).	CO1	K3	8
		(OR)			
2	a)	Let X and Y have joint density function $f(x, y) = \begin{cases} 2e^{-(x+y)} & \text{for } x \geq 0; y \geq 0 \\ 0 & \text{otherwise} \end{cases}$ Then find conditional expectation of (i) Y on X (ii) X on Y	CO2	K1	8
	b)	Explain about Chehychey's Inequality and the utility in statistics	CO2	K3	7
		UNIT - II			
3	a)	It has been claimed that in 60% of all solar installations, 'utility bill reduced to by one- third. Identify the probabilities for the utility bill reduce by at least one- third (i) in four of five installations and (ii) at least four of five installations	CO2	K3	8
	b)	Utilize probability mass function of Poisson's distribution to determine its mean, variance, coefficient skewness & kurtosis.	CO2	K3	7
		(OR)			
4	a)	If 20% of memory chips made in a certain plant are defective, then identify the probabilities, that a randomly chosen 100 chips for inspection (i) at most 15 will defective (ii) at least 25 will be defective (iii) in between 16 and 23 will be defective	CO2	K3	8
	b)	Make use of pdf of the Exponential distribution to find its mean and variance	CO2	K3	7

		UNIT - III																					
5	a)	The following table shows corresponding values of three variables X, Y,Z . Model the least square regression equation $Z = a + bx + cy$ <table border="1"> <tr> <td>x</td><td>1</td><td>2</td><td>1</td><td>2</td><td>3</td></tr> <tr> <td>y</td><td>2</td><td>3</td><td>1</td><td>1</td><td>2</td></tr> <tr> <td>z</td><td>12</td><td>19</td><td>8</td><td>11</td><td>18</td></tr> </table>	x	1	2	1	2	3	y	2	3	1	1	2	z	12	19	8	11	18	CO4	K3	8
x	1	2	1	2	3																		
y	2	3	1	1	2																		
z	12	19	8	11	18																		
	b)	Explain the procedure for fitting an exponential curve of the form $y = a e^{bx}$.	CO4	K5	7																		
		(OR)																					
6	a)	What the properties of a good estimator. Explain each of them	CO3	K1	7																		
	b)	Suppose that n observations $X_1, X_2, \dots, X_n$ are made from normal distribution and variance is unknown. Identify the maximum likelihood estimate of the mean.	CO3	K3	8																		
		UNIT - IV																					
7	a)	Show that in any non- directed graph there is even number of vertices of odd degree.	CO4	K1	8																		
	b)	State and prove Euler's formula for planar graphs	CO4	K2	7																		
		(OR)			7																		
8	a)	Prove that a tree with 'n' vertices have 'n-1' edges	CO4	K3	7																		
	b)	If T is a binary tree of n vertices, show that the number of pendant vertices is $\frac{(n+1)}{2}$	CO4	K1	8																		
		UNIT - V																					
9	a)	Make use of the principles of Inclusion and exclusion find the number of integers between 1 and 100 that are divisible by 2, 3 or 5	CO5	K3	7																		
	b)	Identify the number of integral solutions for $x_1 + x_2 + x_3 + x_4 + x_5 = 50$ where $x_1 \geq 4, x_2 \geq 7, x_3 \geq 14, x_4 \geq 10, x_5 \geq 0$	CO5	K3	8																		
		(OR)																					
10	a)	Solve the recurrence relation $a_n - 7a_{n-1} + 12a_{n-2} = 0$ for $n \geq 2$ using Generating function method.	CO5	K3	8																		
	b)	Solve $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$ for $n \geq 2$.	CO5	K3	7																		

[M19 IT 1102]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
ADVANCED DATA STRUCTURES
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Identify the basic requirements to achieve good Hashing mechanism.	1	3	8
	b).	Classify different collision resolution techniques	1	4	7
		OR			
2.	a).	Apply Following elements 76, 40, 48, 05, 55 to inserted into an empty hash table with hash function $f(x) = x \% 7$ for quadratic probing.	1	3	8
	b).	Distinguish the importance of Double Hashing with example.	1	4	7
		UNIT - II			
3.	a).	Analyze deterministic Skip Lists	2	4	7
	b).	Identify Update Operations on Skip Lists.	1	3	8
		OR			
4.	a).	Identify Search operations on Skip lists.	1	3	7
	b).	Analyze Probabilistic Analysis of Skip Lists.	1	4	8
		UNIT - III			
5.	a).	Construct Binary Search tree with values 13, 3, 4, 12, 14, 10, 5, 1, 8, 2, 7, 9, 11, 6, 18	4	4	8
	b).	Construct Red Black tree algorithm with example	4	4	7
		OR			
6.	a).	Construct a AVL trees, using the result of inserting values 3 , 1 , 4 , 6 , 9 , 2 , 5 , 7 into an initially empty AVL tree?	4	4	8
	b).	Compare B-trees and 2-3 Trees with example	2	4	7
		UNIT - IV			
7.	a).	Analyze Boyer-Moore Algorithm with some example	3	4	8
	b).	Analyze Longest Common Subsequence Problem(LCS)	3	4	7
		OR			
8.	a).	List the advantages and disadvantages of Tries	3	4	7
	b).	Analyze Brute-Force Pattern Matching	3	4	8
		UNIT - V			
9.	a).	Apply Two Dimensional Range Searching	4	3	7
	b).	Construct a Priority Search Tree with example	4	4	8
		OR			
10.	a).	Identify Recent trends in Hashing and Trees	4	3	8
	b).	Construct K-D Trees with example	4	4	7

[M19 IT 1103]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
ARTIFICIAL INTELLIGENCE
Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Construct the state space graph model for vehicle not starting problem	1	4	7
	b).	Apply heuristic search algorithm to find the water jug problem	1	3	8
		OR			
2.	a).	Apply problem reduction strategy for problem solving	1	3	8
	b).	Explain the various problem characteristics of AI	1	2	7
		UNIT - II			
3.	a).	Solve the given propositional calculus expressions are equivalent or not ($P \rightarrow Q \rightarrow R$) AND ($P \rightarrow Q \wedge Q \rightarrow R$)	2	3	9
	b).	Analyze the given expression is tautology ($P \wedge Q \wedge R$) $\vee \sim P$	2	4	6
		OR			
4.	a).	Illustrate the unification algorithm with example	2	2	6
	b).	Apply resolution refutation in propositional logic for checking the equivalence of expressions	2	3	9
		UNIT - III			
5.	a).	Construct semantic network for the mobile device	3	4	7
	b).	Construct the script for patient visiting the hospital	3	4	8
		OR			
6.	a).	Draw & explain the components of Expert system architecture	3	2	7
	b).	Build a rule based expert system for criminal identification.	3	3	8
		UNIT - IV			
7.	a).	Design Bayesian belief network for classification using rain prediction data set	4	4	6
	b).	How can we use the Dempster Shafer theory for prediction?	4	3	9
		OR			
8.	a).	Identify the operations performed on fuzzy set	4	3	7
	b).	Write about different types of membership functions	4	2	8
		UNIT - V			
9.	a).	How the support vector machines can be used in machine learning?	5	3	9
	b).	Differentiate the supervised & unsupervised learning	5	3	6
		OR			
10.	a).	Design a Perceptron for EX-OR gate logic	5	4	8
	b).	Draw the structure of multi layered forward networks	5	2	7

[M19IT1104]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
SERVICE ORIENTED ARCHITECTURES AND WEB SECURITY
(Information Technology)
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT – I			
1.	a).	Explain SAX based parsing with example.	CO1	K2	7
	b).	Give a brief note on Modelling database in XML.	CO1	K3	8
		OR			
2.	a).	Distinguish DOM and SAX based XML parsing.	CO1	K4	7
	b).	Build a document type definition that defines the structure for email message, further create a XML document that reference to the created document type definitions.	CO1	K3	8
		UNIT – II			
3.	a).	Identify the importance of messaging with SOAP and message exchange patterns in web services.	CO2	K3	7
	b).	Explain entity centric and Task centric business service design in detail.	CO2	K2	8
		OR			
4.	a).	Explain briefly about Anatomy of SOA.	CO2	K2	7
	b).	List out different issues involved in principles of service orientation	CO2	K2	8
		UNIT – III			
5.	a).	Summarize about WSDL & SOAP basics in service oriented design.	CO2	K2	7
	b).	Examine how service oriented architecture achieved in .NET platform.	CO2	K4	8
		OR			
6.	a).	Examinehow SOA supports J2EE.	CO2	K4	7
	b).	Analyze the role of UDDI in Service Discovery.	CO2	K4	8
		UNIT – IV			
7.	a).	Explain about JAXB architecture in detail.	CO3	K2	7
	b).	Compare different web service interoperability technologies.	CO3	K4	8
		OR			
8.	a).	Explain about WS-BPEL language basics in detail.	CO3	K2	7
	b).	Analyse about WS-choreography model.	CO3	K4	8
		UNIT – V			
9.	a).	Explain about WS-security language basics in detail.	CO3	K2	7
	b).	Explain about XKMS Structure.	CO3	K2	8
		OR			
10.	a).	Explain about XML Security Overview.	CO3	K2	7
	b).	Explain about XACML in Web Security Service.	CO3	K2	8

[M19 IT 1105]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
INTERNET OF THINGS
(Information Technology)
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Compare Different types of IoT Architectures.	CO1	KL4	7.5
	b).	What are Functional Blocks of an IoT Eco System? Explain.	CO1	KL5	7.5
		OR			
2.	a).	List different Enabling Technologies of IoT.	CO1	KL4	7.5
	b).	Evaluate different phases of Internet of Things.	CO1	KL5	7.5
		UNIT - II			
3.	a).	Examine Application Layer Protocols in IoT.	CO2	KL4	7.5
	b).	What is the Influence of Low Power and Lossy Networks over Routing	CO2	KL5	7.5
		OR			
4.	a).	What are Application Transport Methods? Explain.	CO2	KL5	7.5
	b).	Distinguish Constrained Nodes and Constrained Networks	CO2	KL4	7.5
		UNIT - III			
5.	a).	Discuss about System on Chips.	CO3	KL6	7.5
	b).	Explain Interfaces and Raspberry Pi with Python Programming.	CO3	KL5	7.5
		OR			
6.	a).	Discuss about Design Methodology of IoT	CO3	KL6	7.5
	b).	What is the Influence of Microcontroller in IoT	CO3	KL5	7.5
		UNIT - IV			
7.	a).	Differentiate Structured and Unstructured data, Data in Motion and Data in Rest	CO4	KL4	7.5
	b).	Discuss Briefly about Hadoop Eco-System.	CO4	KL6	7.5
		OR			
8.	a).	Explain about Edge Streaming Analytics and Network Analytics.	CO4	KL5	7.5
	b).	Compare Different types of Python Web Applications.	CO4	KL4	7.5
		UNIT - V			
9.	a).	Discuss about Converged Plant wide Ethernet Model (CPwE)	CO5	KL6	7.5
	b).	Explain about Smart Lighting, Smart Parking and Smart Traffic Control.	CO5	KL5	7.5
		OR			
10.	a).	Briefly Discuss about Power Utility Industry	CO5	KL6	7.5
	b).	Explain about different IoT platforms	CO5	KL5	7.5

[M19 IT 1106]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
OPTIMIZATION TECHNIQUES
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Classify and explain various types of optimization problems with examples.	1	4	8
	b).	Identify the necessary & sufficient conditions for multivariable optimization problem without constraints.	1	3	7
		OR			
2.	a).	solve the maximum or minimum of the function $f(x) = x_1^2 + x_2^2 + x_3^2 - 4x_1 - 8x_2 - 12x_3 + 56$	1	4	8
	b).	Distinguish the gradient of the function and its importance in optimization.	1	4	7
		UNIT - II			
3.	a).	Identify transportation problem and represent it mathematically.	2	3	7
	b).	Min $f(x) = x_1^2 - x_1x_2 + 3x_2^2$. Starting point (1,2) by using steepest descent method. Solve calculations for two cycles.	1	4	8
		OR			
4.	a).	Solve the following non-LPP by Lagrangian multiplier method: Min $Z = 4x_1^2 + 2x_2^2 + x_3 - 4x_1x_2$; st $x_1 + x_2 + x_3 = 15$, $2x_1 - x_2 + 2x_3 = 20$ and $x_i \geq 0 \forall i$	2	4	7
	b).	Identify the Kuhn-Tucker conditions min cost flow problem	2	3	8
		UNIT - III			
5.	a).	Compare single server and multiple server models	2	4	8
	b).	List out Probabilistic inventory control models	2	4	7
		OR			
6.	a).	Classify the terminologies involved in dynamic programming.	2	4	8
	b).	Identify the importance of gradient methods.	2	3	7
		UNIT - IV			
7.	a).	Classify the characteristics of a constrained non-linear programming problem.	3	4	8
	b).	Identify the suitable examples for design constraints and objective function	3	3	7
		OR			
8.	a).	Analyze Greedy algorithm with example	3	4	7
	b).	Prove that a graph of n vertices is a complete graph if its chromatic polynomials $P_n(\lambda) = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1)$	4	5	8
		UNIT - V			
9.	a).	Identify necessary and sufficient conditions of non-LPP with single	3	3	7

		equality constraint.			
	b).	A company produces two types of hats. Each hat of first type requires twice as much as labour time as second type. If all hats are of the second type only, the company can produce a total of 500 hats a day. The market limits daily sales of the first and second type to 150 and 250 hats. Assuming that the profits per hat are Rs.8 for type A and Rs.5 for type B, In the stated problem solve: a) Design Vector b) Objective Function	3	4	8
		OR			
10.	a).	Classify balanced transportation problem.	4	4	8
	b).	List out various methods for finding an initial basic feasible solution for a transportation problem.	4	4	7

[M19 IT 1107]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
PARALLEL COMPUTER ARCHITECTURE
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Distinguish parallel computing with example.	1	4	8
	b).	Analyze one parallel algorithm with examples.	1	4	7
		OR			
2.	a).	Classify search operation in a sorted sequence?	1	4	8
	b).	Distinguish Quantitative Principles of Computer Design	1	4	7
		UNIT - II			
3.	a).	Identify Basic Compiler Techniques for Exposing ILP	2	4	7
	b).	Develop an algorithm to compute the total of n numbers.	1	3	8
		OR			
4.	a).	Compare Mesh transpose and shuffle transpose.	1	4	7
	b).	Develop algorithm for Mesh matrix multiplication.	1	3	8
		UNIT - III			
5.	a).	Develop algorithm for linear array multiplication	2	3	8
	b).	Classify system of linear equations in a parallel way?	2	4	7
		OR			
6.	a).	Analyze algorithm for finding the roots of non-linear equations.	2	4	8
	b).	List out limitations of ILP	2	4	7
		UNIT - IV			
7.	a).	Identify how to compute the connectivity matrix in a graph?	3	3	8
	b).	Develop an algorithm to find all pairs shortest paths of a graph.	3	3	7
		OR			
8.	a).	Classify MIMD alpha beta algorithm.	3	4	7
	b).	List out Ten Advanced Optimizations of Cache Performance	3	4	8
		UNIT - V			
9.	a).	Identify real fault and failures in storage system	3	3	7
	b).	Distinguish Internet archive cluster	3	4	8
		OR			
10.	a).	Classify INTEL software development tools	3	4	8
	b).	List out various simulators	3	4	7

[M19 IT 1108]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
BIG DATA ANALYTICS
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Implement Simple Java Program using Generic Types. Explain Generic Types.	1	3	8
	b).	Differentiate Stacks and Queues.	1	2	7
		OR			
2.	a).	Explain the concept of serialization.	1	2	8
	b).	Discuss about Generic Variables with examples	1	2	7
		UNIT - II			
3.	a).	Explain about Google File System	2	2	7
	b).	Explain the architecture of Building blocks of Hadoop.	2	2	8
		OR			
4.	a).	Mention the configuration of Hadoop Cluster using Fully Distributed Mode in detail	2	2	8
	b).	Name different configuration files in Hadoop?	2	2	7
		UNIT - III			
5.	a).	Explain how matrix multiplication is carried out using Map Reduce algorithm	3	2	8
	b).	Write about Driver Code in Hadoop MapReduce.	3	2	7
		OR			
6.	a).	Describe the old and new Java MapReduce APIs	3	2	8
	b).	How to specify a combiner function? Write a program on application to find the maximum temperature, using a combiner function for efficiency.	3	2	7
		UNIT - IV			
7.	a).	Explain about the Writable wrappers for Java primitives.	4	2	8
	b).	What is writable comparable interface?	4	2	7
		OR			
8.	a).	Write briefly about Writable Concepts i. Text ii. Bytes writable	4	2	7
	b).	Explain about the implementation of raw comparator and custom raw comparator with an example.	4	2	8
		UNIT - V			
9.	a).	Write a brief notes on distributed modes of running PIG Scripts.	5	2	8

	b).	How to create and Manage the database and tables using Hive	5	2	7
		OR			
10.	a).	Illustrate the Architecture of PIG	5	2	8
	b).	Explain how the Data Manipulation Language Works.	5	2	7

CO: Course Outcome

KL: Knowledge Level

M: Marks

[M19 IT 1109]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
PRINCIPLES OF CRYPTOGRAPHY
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Identify the classical cryptosystems and its types	1	3	8
	b).	Compare Euler's and Chinese Remainder theorem	1	4	7
		OR			
2.	a).	Distinguish single round of DES Algorithm and key discarding process of DES	2	4	8
	b).	Analyze Cauchy 's theorem with example	2	4	7
		UNIT - II			
3.	a).	Distinguish max-flow min-cut theorem	2	4	7
	b).	Apply encryption and decryption for RSA algorithm parameters :p=3, q=11, e=7, d=?, M=5.	2	3	8
		OR			
4.	a).	Identify different types of attacks	1	3	7
	b).	Compare SSL and SET protocols	2	4	8
		UNIT - III			
5.	a).	Distinguish Diffie Hellman key Exchange algorithm with merits and demerits	2	4	8
	b).	Classify Modes of operation in Block cipher	2	4	7
		OR			
6.	a).	Identify different modes of operations in DES	2	3	8
	b).	Identify different Phases, types of virus structure and types of viruses	1	3	7
		UNIT - IV			
7.	a).	Classify four types of block cipher design	2	4	8
	b).	Analyze the key generation in AES algorithm and its expansion format	3	4	7
		OR			
8.	a).	List out the properties of hash function in cryptography	3	4	7
	b).	Analyze digital signature with ElGamal public key cryptosystem	3	4	8
		UNIT - V			
9.	a).	Distinguish secure hash Algorithm (SHA)	3	4	7
	b).	Compare any two classical crypto systems with suitable examples	1	4	8
		OR			
10.	a).	Classify RSA digital signature	3	4	8
	b).	Compare stream cipher with RC4	3	4	7

[M19IT1110]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
CLUSTER AND GRID COMPUTING
(Information Technology)
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Identify different issues in Grid Computing architecture.	CO1	K3	7
	b).	What are differences between Web services and Service Oriented Architectures.	CO1	K2	8
		OR			
2.	a).	Develop a model of Service description for Web Services.	CO1	K3	7
	b).	Illustrate functional requirements of Grid Computing?	CO1	K2	8
		UNIT - II			
3.	a).	Demonstrate about the basic services offered by OGSA.	CO2	K2	7
	b).	What are data centres and what is their necessary? Identify the functional requirements of OGSA.	CO2	K3	8
		OR			
4.	a).	Build a model to integrate of Databases with the Grid?	CO2	K3	7
	b).	Explain about GT4 Architecture.	CO2	K2	8
		UNIT - III			
5.	a).	Identify programming skeletons for parallel programming.	CO2	K3	7
	b).	Explain about the levels and layers of Single System Image.	CO2	K2	8
		OR			
6.	a).	Analyze the strategies for developing parallel applications.	CO2	K4	7
	b).	Illustrate how the Resource Management is done in Cluster Computing?	CO2	K3	8
		UNIT - IV			
7.	a).	Explain about Global Clocks Synchronization.	CO3	K2	7
	b).	Build a setup of simple cluster and setting up of nodes.	CO3	K3	8
		OR			
8.	a).	Build to Configurations of high availability in Cluster Computing?	CO3	K3	7
	b).	Identify the types of failures and errors in Cluster Computing.	CO3	K3	8
		UNIT - V			
9.	a).	Organize different Scheduling policies in Cluster Computing.	CO3	K3	7
	b).	Identify Modelling Parameters of Load Balancing.	CO3	K3	8
		OR			
10.	a).	Choose any two technologies in Cluster and Grid Computing.	CO3	K3	7
	b).	Build Job Management System in Cluster Computing.	CO3	K3	8

CO: Course Outcome

KL: Knowledge Level

M: Marks

[M19IT1111]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
IMAGING AND MULTIMEDIA SYSTEMS
(Information Technology)
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Apply the image Processing techniques for storing handling the images	1	3	10
	b).	Write about 2 colour models of any	1	2	5
		OR			
2.	a).	Define multimedia & various Multimedia Elements Discuss about	1	2	10
	b).	Design a Multimedia application for online assignment	1	4	5
		UNIT - II			
3.	a).	Discuss about JPEG File Format	1	2	9
	b).	How can we apply Redundancy reduction scheme for image compression	1	3	6
		OR			
4.	a).	Differentiate Rich Text Format & TIFF File Format	2	3	9
	b).	Discuss about Audio Compression techniques	2	2	6
		UNIT - III			
5.	a).	Discuss about Thresholding & convolution in 2D image processing	2	3	7
	b).	Differentiate Steganography & Cryptography			
		OR			
6.	a).	Discuss about Chaos based and Non-Chaos based cryptography methods.	2	2	10
	b).	How can we apply Noise Reduction techniques in Image	2	3	5
		UNIT - IV			
7.	a).	Discuss about Multimedia I/O Technologies	2	2	10
	b).	How Animation is implemented for real time applications	2	3	5
		OR			
8.	a).	How the video is recorded by digital camera	3	3	10
	b).	How Digital Voice is recorded	3	3	5
		UNIT - V			
9.	a).	Differentiate various Multimedia Systems	3	3	10
	b).	Draw the Components of Multimedia Systems & functionalities	2	2	5
		OR			
10.	a).	-Multimedia Authoring Systems – Multimedia Authoring Tools -User Interface	3	2	8
	b).	Discuss about Hypermedia Message Components &Hypermedia Linking	3	2	7

CO: Course Outcome

KL: Knowledge Level

M: Marks

[M19 IT 1112]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech I Semester (R19) Regular Examinations
ADVANCED GRAPH THEORY
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Analyze the maximum number of edges in a simple graph with n vertices is $n(n-1)/2$.	1	4	8
	b).	Prove that if a graph has exactly two vertices of odd degree, there must be path joining these two vertices.	1	5	7
		OR			
2.	a).	Identify Hamiltonian path as spanning tree	1	3	8
	b).	List out some of the properties of tree.	1	4	7
		UNIT - II			
3.	a).	Distinguish max-flow min-cut theorem	2	4	7
	b).	Prove that in any tree, there are at least two pendant vertices	1	5	8
		OR			
4.	a).	Compare Fundamental cut set and Fundamental circuit in a graph.	1	4	7
	b).	List out some types of digraph with suitable example	1	4	8
		UNIT - III			
5.	a).	Prove that every connected graph has at least one spanning tree	2	5	8
	b).	Distinguish Tutte's f - factor theorem with example	2	4	7
		OR			
6.	a).	Identify problems in Euler digraph	2	3	8
	b).	Distinguish Turen's with Example.	2	4	7
		UNIT - IV			
7.	a).	Solve the recurrence relation. $6a_n - 7a_{n-1} = 0, n \geq 1, a_3 = 343$.	3	3	8
	b).	Analyze Traveling Salesman Problem	3	4	7
		OR			
8.	a).	Analyze Greedy algorithm with example	3	4	7
	b).	Prove that a graph of n vertices is a complete graph if its chromatic polynomial is $P_n(\lambda) = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1)$	3	5	8
		UNIT - V			
9.	a).	Compare direct paths and cycle	3	4	7
	b).	Distinguish max flow min cut theorem	3	4	8
		OR			
10.	a).	Classify Brooks theorem with example	3	4	8
	b).	Identify greedy algorithm	3	3	7

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
[M19RD1101]

I M. Tech I Semester (R19) Regular Examinations
RESEARCH METHODOLOGY AND IPR
Common to CST,CS,PSA,IT & CAD
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

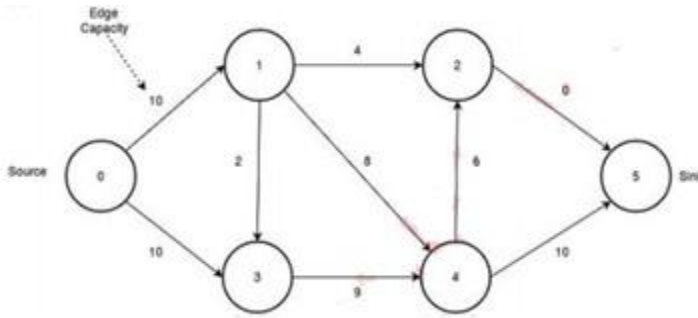
			CO	KL	M
		UNIT-I			
1.	a).	Write briefly about good Research criteria.	CO1	K2	8
	b).	What are the errors in selecting a research problem?	CO1	K2	7
		OR			
2.		Describe briefly the Research process with neat sketch.	CO1	K2	15
		UNIT-II			
3.	a).	Write briefly about Effective Literature studies approaches.	CO1	K2	8
	b).	Explain about Research ethics.	CO3	K2	7
		OR			
4.	a).	Write briefly about Effective technical writing.	CO2	K3	8
	b).	Explain about the Format of research proposal.	CO2	K3	7
		UNIT-III			
5.		Write about the various steps in acquisition of trademarks rights.	CO5	K2	15
		OR			
6.	a).	Write briefly about International cooperation on Intellectual Property.	CO5	K2	8
	b).	Explain the procedure for grants of patents.	CO5	K2	7
		UNIT-IV			
7.		Explain about patent information and databases.	CO5	K2	15
		OR			
8.	a).	Write briefly about scope of patent rights.	CO5	K2	8
	b).	Write briefly about Licensing and transfer of technology.	CO4	K2	7
		UNIT-V			
9.		Write briefly about Administration in patent system.	CO5	K2	15
		OR			
10.		Write briefly about New developments in IPR.	CO6	K2	15
CO-COURSE OUTCOME			KL-KNOWLEDGE LEVEL		M-MARKS

[M19 IT 1201]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
ADVANCED ALGORITHMS
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M																																				
UNIT - I																																									
1.	a).	Apply Quick sort algorithm in the following data and comment on complexity 40, 10, 15, 20, 35, 45, 25, 15, 45, 10	1	3	8																																				
	b).	Analyze the performance of an Binary search and linear search algorithms. Compute complexity of binary search and linear search	1	4	7																																				
OR																																									
2.	a).	Evaluate the maximum flow for the following graph 	2	5	8																																				
	b).	Apply BFS Algorithm for the following values Given graph with 5 nodes and 3 edges, [1,2], [1,3], [3,4]	1	4	7																																				
UNIT - II																																									
3.	a).	Solve the following activity selection problem with greedy algorithm? <table border="1" data-bbox="302 1486 1261 1593"><tr><td>i</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>S_i</td><td>1</td><td>3</td><td>0</td><td>5</td><td>3</td><td>5</td><td>6</td><td>8</td><td>8</td><td>2</td><td>12</td></tr><tr><td>f_i</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr></table>	i	1	2	3	4	5	6	7	8	9	10	11	S _i	1	3	0	5	3	5	6	8	8	2	12	f _i	4	5	6	7	8	9	10	11	12	13	14	2	3	8
i	1	2	3	4	5	6	7	8	9	10	11																														
S _i	1	3	0	5	3	5	6	8	8	2	12																														
f _i	4	5	6	7	8	9	10	11	12	13	14																														
	b).	Classify greedy method by taking suitable example of actively selection problem	2	3	7																																				
OR																																									
4.	a).	Classify characterization of maximum matching by augmenting paths	1	3	7																																				
	b).	Apply Dijkstra’s algorithm for single source shortest paths. How can it be used for solving all pairs shortest paths problem?	2	3	8																																				
UNIT - III																																									
5.	a).	Analyze the maximum flow using Ford Flukerson Algorithm	2	4	8																																				
	b).	Solve Edmond-Karp maximum-flow algorithm with example	2	3	7																																				

		OR			
6.	a).	Apply divide and conquer method to solve maximum sub array problem	2	3	8
	b).	Analyze max flow-mincut theorem with an example	1	4	7
		UNIT - IV			
7.	a).	Apply Chinese remainder theorem, to the following equations: $a = 2 \pmod{5}$ $b = 3 \pmod{13}$ Generate solutions in the form of table	2	3	8
	b).	Distinguish polynomial multiplication and division	1	4	7
		OR			
8.	a).	Identify Conversions between base-representation and modulo-representation	1	3	7
	b).	Evaluate Fast Fourier Transform algorithm with an example	3	5	8
		UNIT - V			
9.	a).	Prove that clique problem is NP-complete.	3	4	7
	b).	Prove Travelling Salesman Problem as NP complete.	3	4	8
		OR			
10.	a).	Prove that feedback edge set problem in NP-complete.	3	4	8
	b).	Compare polynomial multiplication and division	3	4	7

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19IT1202]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
FULL STACK TECHNOLOGIES
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a).	Compare HTML and XHTML.	8	CO1	L4
	b).	Analyze HTTP and WWW	7	CO1	L4
		OR			
2.		Design a form using HTML. (Use minimum 15 Different tags)	15	CO1	L6
		UNIT-II			
3.	a).	Analyze Style Rule Cascading and Inheritance in CSS.	8	CO2	L4
	b).	Explain about Boot strap in CSS3	7	CO2	L5
		OR			
4.	a).	Discuss about JScripts objects.	8	CO2	L5
	b).	Analyze Control Statements in Javascript.	7	CO2	L5
		UNIT-III			
5.		Design a web form to insert and retrieve data using JSP.	15	CO3	L6
		OR			
6.	a).	Analyze JavaBeans Classes.	8	CO3	L5
	b).	Explain about JQuery.	7	CO3	L5
		UNIT-IV			
7.	a).	Creating First Controller using AngularJS	8	CO3	L5
	b).	Explain about Unit Testing in AngularJS.	7	CO3	L5
		OR			
8.	a).	Explain about AngularJS Modules.	8	CO4	L5
	b).	Explain about Error Handling with Forms.	7	CO4	L5
		UNIT-V			
9.	a).	Explain about React Components.	8	CO4	L5
	b).	Explain about ReactDOM.	7	CO4	L5
		OR			
10.	a).	Compare Obstacles and Roadblocks.	8	CO4	L5
	b).	Explain about DOM Rendering.	7	CO4	L5

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19IT1203]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

I M. Tech II Semester (R19) Regular Examinations
MACHINE LEARNING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a)	Compare Structured and Unstructured, Forms of Learning.	8	CO1	L4
	b)	Explain about Productive use of Machine Learning	7	CO1	L4
		OR			
2.	a)	Analyze various Applications in diverse fields	8	CO1	L4
	b)	Explain about Data Representation	7	CO1	L4
		UNIT-II			
3.		Compare Metrics for assessing regression and classification.	15	CO2	L4
		OR			
4.		Discuss about Computational Learning Theory and Occam's Razor Principle.	15	CO2	L5
		UNIT-III			
5.	a)	Discuss about K-Nearest Neighbor Classifier.	8	CO3	L5
	b)	Compare Linear Regression and Logistic Regression.	7	CO3	L5
		OR			
6.	a)	Explain about Fisher's Linear Discriminant.	8	CO3	L5
	b)	Explain about Minimum Description Length Principle.	7	CO3	L5
		UNIT-IV			
7.	a)	Discuss about Regression by Support vector Machines	15	CO4	L6
	b)	Explain about Widrow-Hoff Learning Rule in Neural Networks.			
		OR			
8.	a)	Explain about Neuron Models in Neural Networks.	8	CO4	L5
	b)	Discuss about Perceptron Algorithm in SVM.	7	CO4	L5
		UNIT-V			
9.		Explain about Decision Tree Learning of Machine Learning.	15	CO5	L5
		OR			
10.	a)	Explain about error back propagation algorithm.	8	CO5	L5
	b)	Discuss about Multilayer Perceptron Networks.	7	CO5	L5

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19IT1204]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
DevOps
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.		Analyze the Phases of Software Development life cycle	15	CO1	L4
		OR			
2.		Analyze the principles of agile software development	15	CO1	L4
		UNIT-II			
3.		Analyze Fundamentals of DevOps.	15	CO2	L4
		OR			
4.	a).	Discuss about Architecture of DevOps.	8	CO2	L5
	b).	Analyze DevOps eco system.	7	CO2	L4
		UNIT-III			
5.		Analyze DevOps adoption in projects	15	CO2	L6
		OR			
6.	a).	Analyze Agiling capabilities in DevOps.	8	CO2	L5
	b).	Explain about Tool stack implementation in DevOps.	7	CO2	L5
		UNIT-IV			
7.	a).	Explain about Continuous Integration of CI/CD	8	CO5	L5
	b).	Explain about Metrics to track CICD practices.	7	CO5	L5
		OR			
8.	a).	Explain about Benefits of CI/CD.	8	CO5	L5
	b).	Explain about Continuous Delivery and Deployment of CI/CD.	7	CO5	L5
		UNIT-V			
9.		Discuss about Devops Maturity Model	15	CO4	L5
		OR			
10.	a).	List out stages of Devops maturity model.	8	CO4	L5
	b).	Explain about Key factors of DevOps maturity model.	7	CO4	L5

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19 IT 1205]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
ADVANCED NETWORK PRINCIPLES AND PROTOCOLS
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Analyze Topology switching architecture with neat diagram	1	3	8
	b).	Identify various classes of IEEE 802.X Standard Ethernet.	1	4	7
		OR			
2.	a).	Compare ALOHA and CSMA/CD Protocols with an example?	2	4	8
	b).	List out various Applications of Networks	2	4	7
		UNIT - II			
3.	a).	Identify the different features in Transmission control protocol	2	4	7
	b).	Analyze Flow control and Congestion control in TCP	2	3	8
		OR			
4.	a).	Analyze Spanning tree protocol with neat diagram	1	4	7
	b).	Distinguish Input and output modules with example	2	4	8
		UNIT - III			
5.	a).	Compare ARP with RARP with neat diagram	2	3	8
	b).	Identify Various issues addressed by transport layer protocol	2	4	7
		OR			
6.	a).	Identify the operations of Dynamic Host Configuration Protocol	2	3	8
	b).	Classify the header formats for the three major protocols in the TCP/IP protocol suite.	1	4	7
		UNIT - IV			
7.	a).	Compare Transmission Control Protocol with User Datagram Protocol	2	4	8
	b).	Identify the User Datagram Protocol services.	3	4	7
		OR			
8.	a).	List out various typical applications of User Datagram Protocol	3	4	7
	b).	Identify Various issues addressed by transport layer protocol	3	3	8
		UNIT - V			
9.	a).	Distinguish about Simple Network Management Protocol with neat diagram	3	4	7
	b).	Identify the operations of Bootstrap Network Time Protocol	1	3	8
		OR			
10.	a).	Identify the characteristics of Simple Mail Transfer Protocol	3	3	8
	b).	Distinguish about Trivial File Transfer Protocol	3	4	7

M-MARKS
LEVEL

CO-COURSE OUTCOME

KL-KNOWLEDGE

[M19IT1206]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

I M. Tech II Semester (R19) Regular Examinations

DISTRIBUTED COMPUTING

Department of Information Technology

MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer ONE Question from EACH UNIT.

All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a)	Compare Message-passing systems and shared memory systems.	8	CO1	L5
	b)	List out Models of process communications	7	CO1	L4
		OR			
2.	a)	Compare Synchronous versus asynchronous executions	8	CO1	L5
	b)	List out Models of communication networks	7	CO1	L4
		UNIT-II			
3.	a)	Explain about Group communication.	8	CO2	L5
	b)	Discuss about Message ordering paradigms.	7	CO2	L5
		OR			
4.	a)	Explain about Total order.	8	CO2	L5
	b)	Discuss about Snapshot algorithms for FIFO channels.	7	CO2	L5
		UNIT-III			
5.	a)	Discuss about Lamport's algorithm.	8	CO3	L5
	b)	Explain about Knapp's classification.	7	CO3	L5
		OR			
6.	a)	Explain about Maekawa's algorithm.	8	CO3	L5
	b)	Explain about Algorithms for the single resource model.	7	CO3	L5
		UNIT-IV			
7.	a)	Discuss about Issues in failure recovery.	8	CO4	L5
	b)	Explain about Agreement in a failure –free system.	7	CO4	L5
		OR			
8.	a)	Discuss about Log-based rollback recovery.	8	CO4	L5
	b)	Explain about Agreement in synchronous systems with failures.	7	CO4	L5
		UNIT-V			
9.		Explain about Peer-to-peer computing Data indexing and overlays.	15	CO5	L5
		OR			
10.	a)	List out Memory consistency models.	8	CO5	L5
	b)	Discuss about Shared memory Mutual Exclusion.	7	CO5	L5

**M-MARKS
LEVEL**

CO-COURSE OUTCOME

KL-KNOWLEDGE

[M19IT1207]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

I M. Tech II Semester (R19) Regular Examinations
SOCIAL NETWORK ANALYTICS
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a).	Explain about Preliminaries and definitions of Social Networks.	8	CO1	L5
	b).	Explain about Centrality measures of Social Networks.	7	CO1	L5
		OR			
2.	a).	Explain about Balance and Homophily of Social Networks.	8	CO1	L5
	b).	Explain about Erdos Number Project of Social Networks.	7	CO1	L5
		UNIT-II			
3.	a).	List out the Models of network growth.	8	CO2	L5
	b).	Analyze the Structural equivalence in random graph model.	7	CO2	L4
		OR			
4.	a).	Discuss about Multidimensional Scaling.	8	CO2	L5
	b).	Analyze Navigation in social Networks.	7	CO2	L4
		UNIT-III			
5.		Discuss about Network topology and diffusion	15	CO2	L5
		OR			
6.	a).	Analyze Contagion in Networks.	8	CO2	L4
	b).	Explain about Complex contagion.	7	CO2	L5
		UNIT-IV			
7.	a).	Explain about The ErdosRenyi Model	8	CO5	L5
	b).	Explain about Small world experiments.	7	CO5	L5
		OR			
8.	a).	Explain about origins of small world.	8	CO5	L5
	b).	Explain about Clustering Models.	7	CO5	L5
		UNIT-V			
9.		Discuss about Important vertices and page rank algorithm	15	CO4	L5
		OR			
10.	a).	Analyze biased voting.	8	CO4	L4
	b).	Explain about basics of game theory.	7	CO4	L5

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19 IT 1208]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
DIGITAL IMAGE PROCESSING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Identify the fundamentals steps of digital image processing.	1	3	8
	b).	Distinguish the concepts of sampling and quantization	1	4	7
		OR			
2.	a).	With the neat block diagram, classify the components of general purpose image processing system.	1	4	8
	b).	Classify the image acquisition using single sensor	1	4	7
		UNIT - II			
3.	a).	Classify image smoothing of images using ideal, Butterworth and gaussian low pass filter	2	4	7
	b).	For a given 4 X 4 image having gray scales between [0,9] perform histogram equalization and draw the histogram of image before and after equalization, 4 X 4 image is shown in below 2 3 3 2 4 2 4 3 3 2 3 5 2 4 2 4	2	3	8
		OR			
4.	a).	With necessary graphs, explain the log nd power law transformation used for spatial image enhancement	3	3	7
	b).	Compare two band sub band coding and decoding system	2	4	8
		UNIT - III			
5.	a).	List out various requirements for multi-resolution analysis? explain	3	4	8
	b).	Draw the functional block diagram of image compression system and explain the purpose of each block	3	3	7
		OR			
6.	a).	Analyze the following morphological algorithms a) Boundary extraction b) Hole Filling	5	4	8
	b).	Distinguish wavelet transform in two dimensions	4	4	7
		UNIT - IV			
7.	a).	Identify the different ways to estimate the degradation function.	4	3	8
	b).	Classify image sharpening using laplacian operator	2	3	7
		OR			
8.	a).	Compute the Haar transform of the 2 X 2 image	3	3	7

		$F = \begin{bmatrix} 3 & -1 \\ 6 & 2 \end{bmatrix}$			
	b).	Prove that both the 2-D continuous and discrete Fourier transforms are linear operations.	3	5	8
		UNIT - V			
9.	a).	Analyze edge linking using Hough algorithm	5	4	7
	b).	Distinguish region based segmentation	5	4	8
		OR			
10.	a).	With necessary figures, classify the opening and closing operations.	5	3	8
	b).	Classify Hit-or-Mass transformation technique.	5	3	7

**M-MARKS
LEVEL**

CO-COURSE OUTCOME

KL-KNOWLEDGE

[M19IT1209]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

I M. Tech II Semester (R19) Regular Examinations

BLOCK CHAIN TECHNOLOGIES

Department of Information Technology

MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a).	Explain about Asynchronous Byzantine Agreement.	8	CO1	L5
	b).	Explain about GARY model in block Chain.	7	CO1	L5
		OR			
2.	a).	Compare PoS And PoW models in block Chain.	8	CO1	L5
	b).	Analyze AAP protocol.	7	CO1	L4
		UNIT-II			
3.	a).	Explain the elliptic curve cryptography.	8	CO2	L5
	b).	Analyze the signature schemes.	7	CO2	L4
		OR			
4.	a).	Discuss about Cryptographic basics for cryptocurrency.	15	CO2	L5
		UNIT-III			
5.		Discuss about mathematical analysis of properties of Bitcoin	15	CO2	L5
		OR			
6.	a).	Analyze hardness of mining.	8	CO2	L4
	b).	Explain about double spending.	7	CO2	L5
		UNIT-IV			
7.	a).	Explain about The Ethereum Virtual Machine	8	CO4	L5
	b).	Explain about attacks on smart contracts.	7	CO4	L5
		OR			
8.	a).	Explain about Wallets for Ethereum.	8	CO4	L5
	b).	Explain about Smart Contracts.	7	CO4	L5
		UNIT-V			
9.		Discuss about Zero Knowledge proofs and protocols in Blockchain	15	CO5	L5
		OR			
10.	a).	Analyze pairing on Elliptic curves.	8	CO5	L4
	b).	Explain about SNARK.	7	CO5	L5

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19IT1210]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
DATA SCIENCE
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.

All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a).	Describe data science process in detail.	8	CO1	L3
	b).	Discuss some of the key components of Data Science.	7	CO1	L4
		OR			
2.	a)	Illustrate Data Science toolkit framework with example application.	8	CO1	L4
	b)	Explain Different types of data with suitable examples	7	CO1	L3
		UNIT-II			
3.	a).	Identify the sources of Data and explain about data collection using APIs.	8	CO3	L3
	b).	Analyze and explain about data exploration analysis with tools.	7	CO3	L4
		OR			
4.	a).	Develop a model for Data storage and Management using multiple data sources.	8	CO3	L4
	b).	Discuss about data model. How to create and evaluate a data model. Describe with one case study.	7	CO3	L4
		UNIT-III			
5.	a)	Apply distribution model to predict values using sample data.	7	CO2	L4
	b)	Explain about implementation of simple linear regression model.	8	CO2	L3
		OR			
6.	a).	Explain SVM Algorithm with example.	8	CO2	L4
	b).	Explain classification techniques with suitable example.	7	CO2	L4
		UNIT-IV			
7.	a).	Analyze the role of data visualization in data science.	7	CO4	L4
	b).	Application of encoding techniques in data visualization.	8	CO4	L3
		OR			
8.		Illustrate visual encoding techniques in data science with examples	15	CO4	L4
		UNIT-V			
9.	a).	Differentiate qualitative and quantitative methods in primary data collection	7	CO4	L3
	b)	Illustrate the issues in selecting survey methods	8	CO4	L4
		OR			
10.		Describe the development methods in data science using real time applications	15	CO4	L4

**M-MARKS
LEVEL**

CO-COURSE OUTCOME

KL-KNOWLEDGE

[M19 IT 1211]

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)

I M. Tech II Semester (R19) Regular Examinations

SOFT COMPUTING

Department of Information Technology

MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer ONE Question from EACH UNIT

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Compare and contrast Human brain and Neural network	1	3	8
	b).	Analyze fuzzy set theory with example	1	4	7
		OR			
2.	a).	List out different applications of Fuzzy logic concepts in Engineering problems	1	3	8
	b).	Classify the following- Fuzzy rules and Fuzzy Reasoning	1	4	7
		UNIT - II			
3.	a).	Classify the limitations of Derivative-Free Optimization?	2	4	7
	b).	Analyze the method of steepest descent	2	3	8
		OR			
4.	a).	Distinguish about downhill simplex search	2	4	7
	b).	Identify the features of Derivative based optimization	2	4	8
		UNIT - III			
5.	a).	Identify the techniques for Heuristic Search and explain	3	3	8
	b).	Analyze predicate calculus algorithm	3	4	7
		OR			
6.	a).	Identify the issues and acquisition of Rule based knowledge	3	3	8
	b).	Explain about heuristic classification state space search	3	3	7
		UNIT - IV			
7.	a).	With help of the architecture analyze the Hybrid learning algorithm	4	4	8
	b).	Classify Learning methods that cross fertilize ANFIS and RDFN	4	4	7
		OR			
8.	a).	Analyze the Coactive fuzzy Modeling	4	4	7
	b).	Classify the Neuro Fuzzy Inference systems	4	4	8
		UNIT - V			
9.	a).	Analyze Automobile fuel efficiency prediction	5	4	7
	b).	Classify soft computing for color recipe prediction	5	4	8
		OR			
10.	a).	Explain Inverse Kinematics Problems	5	3	8
	b).	Classify Printed character Recognition	5	4	7

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE

LEVEL

[M19 IT 1212]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
I M. Tech II Semester (R19) Regular Examinations
NATURAL LANGUAGE PROCESSING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Identify the NLP Tasks in Syntax, Semantics and pragmatics	1	3	8
	b).	Distinguish the applications like information extraction, question answering and machine translation	1	4	7
		OR			
2.	a).	Identify the role of machine learning in Natural Language Processing	1	3	8
	b).	Classify the problem of ambiguity with example	1	4	7
		UNIT - II			
3.	a).	Distinguish the role of language models in N-gram Language	2	4	7
	b).	Classify Hidden Markov Models with an example	4	4	8
		UNIT - III			
4.	a).	Compare the estimating parameters and smoothing	2	3	7
	b).	Evaluate any one of the language model with example	2	4	8
		UNIT - IV			
5.	a).	Explain about parsing for context free grammars (CFGs)	5	3	8
	b).	Describe logical form as an intermediate representation in natural language processing with diagram	3	3	7
		OR			
6.	a).	Distinguish Grammar formalisms and tree banks	3	4	8
	b).	Describe lexical probabilities and techniques of obtaining them in natural language processing	3	3	7
		UNIT - V			
7.	a).	Identify the uses of semantic web ontology	4	3	8
	b).	Classify word sense disambiguation in natural language processing	4	4	7
		UNIT - V			
8.	a).	Distinguish lexical resource wordnet used in natural language processing	4	4	7
	b).	Describe Probabilistic parsing in natural language processing	4	3	8
		UNIT - V			
9.	a).	Explain machine translation system evaluation tools	5	3	7
	b).	How is natural language processing useful in an automatic text clustering problem	5	3	8
		OR			
10.	a).	Describe automatic machine translation problem	5	3	8
	b).	Explain Compositional semantic with example	4	3	7

**M-MARKS
LEVEL**

CO-COURSE OUTCOME

KL-KNOWLEDGE

[M19 IT 2101]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
M. Tech III Semester (R19) Regular Examinations
DEEP LEARNING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**
 All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Explain about the fundamental Learning Paradigms of Deep Learning	1	3	8
	b).	Explain the perspectives of deep learning framework	1	4	7
		OR			
2.	a).	Identify the different issues addressed in deep learning frameworks	4	4	8
	b).	what is the role of activation functions in neural network?	3	4	7
		UNIT - II			
3.	a).	What Do you understand by Backpropagation in Neural Networks?	1	4	7
	b).	Explain Greedy layer wise unsupervised pretraining.	3	3	8
		OR			
4.	a).	Analyze optimization techniques in Deep Learning.	4	4	7
	b).	Identify the difficulty of training the Deep Neural Networks.	2	4	8
		UNIT - III			
5.	a).	What is the difference between feedforward neural network and recurrent neural network?	2	3	8
	b).	What are the different layers in CNN? What is pooling in CNN and how does it work?	2	4	7
		OR			
6.	a).	What is Regularization? Why do we need L1 and L2 Regularizations?	1	3	8
	b).	Explain the architecture of Deep Belief Network? List the Applications of Deep Belief Networks.	4	4	7
		UNIT - IV			
7.	a).	Explain Probabilistic neural network model with an example	3	4	8
	b).	Explain about different computational role for Hopfield nets	2	4	7
		OR			
8.	a).	What are autoencoders? Explain different types of autoencoders.	2	4	8
	b).	What are Restricted Boltzmann Machines and compare Autoencoders & RBMs	3	3	7
		UNIT - V			
9.	a).	Explain how you can Video images in computer vision.	3	4	7
	b).	Why tensor flow is most preferred library in Deep Learning?	3	3	8
		OR			
10.	a).	Explain object recognition with real time example.	4	3	8
	b).	List the applications of deep learning in Natural language processing?	4	4	7

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19 IT 2102]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
M. Tech III Semester (R19) Regular Examinations
EMBEDDED COMPUTING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
UNIT - I					
1.	a).	What is an Embedded System?, Embedded system components.	1	2	8
	b).	Discuss Linux-based embedded system Reference hardware model.	1	2	7
OR					
2.	a).	Differentiate Linux Based Embedded system and Embedded system.	1	2	8
	b).	Implementations of Reference hardware model.	1	3	7
UNIT - II			1	2	
3.	a).	Explain Linux architecture.	1	2	7
	b).	Briefly discuss Memory manager.			8
OR					
4.	a).	Discuss Virtual file system architecture.	1	2	7
	b).	What is Inter-process communication and explain Inter-process communication architecture.	1	2	8
UNIT - III					
5.	a).	Compare CPU – I/O interface & CPU – I/O interface with polling.	2	2	7
	b).	Explain Direct memory access (DMA) architecture.	2	2	8
OR					
6.	a).	Briefly discuss Virtual File System (VFS).	2	2	8
	b).	Discuss Linux kernel modules.	2	2	7
UNIT - IV					
7.	a).	Explain Memory Mapping- Page Alignment.	2	2	7
	b).	Differentiate Memory Mappings and Unmapping Regions.	2	3	8
OR					
8.	a).	Explain Syncing Memory Regions to Disk	2	2	7
	b).	Discuss Locking Memory Regions and Types of Locking methods.	2	2	8
UNIT - V					
9.	a).	Briefly discuss Networking with Socket.	3	2	8
	b).	Differentiate IPv4 and IPv6 Addressing	3	3	7
OR					
10.	a).	Discuss TCP/IP protocol.	3	2	8
	b).	Explain Socket operations.	3	2	7

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19 IT 2103]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
M. Tech III Semester (R19) Regular Examinations
ETHICAL HACKING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**.
 All questions carry equal marks.

			M	CO	KL
		UNIT-I			
1.	a)	Define ethical hacking. List its limitations	8	CO1	L3
	b)	Illustrate about foundation for ethical hacking.	7	CO1	L4
		OR			
2		Explain Hacking methodologies in detail	15	CO1	L3
		UNIT-II			
3.		Discuss in detail about Advanced Windows Hacking	15	CO1	L3
		OR			
4.	a)	How the application is hacked.	8	CO1	L3
	b)	Discuss the Operating System Hacking.	7	CO1	L3
		UNIT-III			
5.		How will you crack the password. Elaborate the steps in detail	15	CO1	L4
		OR			
6.		Discuss in detail about i) Cracking Login Password ii) Cracking BIOS password iii) Cracking Unix Password	15	CO1	L4
		UNIT-IV			
7.	a)	Explain the functions associated with functions in PERL	8	CO2	L4
	b)	Differentiate Chomp() and Chop()	7	CO2	L5
		OR			
8.	a)	Discuss about Conditional statements and Looping Statements in PERL	8	CO2	L3
	b)	Write about File handling in PERL	7	CO2	L3
		UNIT-V			
9.		Discuss the procedure for identifying unknown device drivers	15	CO3	L3
		OR			
10.	a)	Explain about Creating new virus strains	8	CO3	L4
	b)	Discuss about working of virus	7	CO3	L3

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

[M19 IT 2104]
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)
M. Tech III Semester (R19) Regular Examinations
DIGITAL MARKETING
Department of Information Technology
MODEL QUESTION PAPER

TIME: 3 Hrs.

Max. Marks: 75 M

Answer **ONE Question** from **EACH UNIT**

All questions carry equal marks

			CO	KL	M
		UNIT - I			
1.	a).	Analyze Dynamic Document content and properties.	1	3	8
	b).	Discuss about web pages with basic HTML5 with example.	1	2	7
		OR			
2.	a).	Distinguish the importance of Scripting Documents.	1	3	8
	b).	Identify various HTML5 Elements and explain each	1	2	7
		UNIT - II			
3.	a).	Explain variables, operators and control structures of objects in Java script.	2	2	7
	b).	Explain the various types of XML schema data types and their applications.	2	2	8
		OR			
4.	a).	Define CSS. Explain inline, internal, external and embedded style sheets with examples.	2	3	8
	b).	Write a java script to find factorial of a given number.	2	2	7
		UNIT - III			
5.	a).	Explain planning and executing a search advertising campaign	3	2	8
	b).	Differentiate Measuring and Mapping of Search Engine.	3	3	7
		OR			
6.	a).	What is Social media? explain social media marketing and its strategy	3	2	8
	b).	Explain Strategic Implications of Advertising on the search network	3	2	7
		UNIT - IV			
7.	a).	Discuss Social Media Marketing Strategy.	4	2	8
	b).	Differentiate Social media marketing and Online marketing.	4	2	7
		OR			
8.	a).	Discuss Forms of Mobile Marketing, Mobile Advertising and M-Commerce.	4	2	8
	b).	Explain Adopting Social Media in Organizations.	4	2	7
		UNIT - V			
9.	a).	Discuss about content marketing in India	4	2	7
	b).	What is Email marketing? Discuss E-Mail Marketing Strategy	4	3	8
		OR			
10.	a).	Discuss Execution of E-Mail Marketing in india	4	3	8
	b).	Discuss Internet marketing Strategies	4	3	7

M-MARKS

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL